

Cole Meyer

Lunar and Planetary Laboratory
1629 E. University Blvd.
Tucson, AZ 85721

<https://colemeyer.github.io/>
cmmeyer@arizona.edu
(320) 296-9560

EDUCATION

University of Arizona, Tucson, AZ, 2024–present

Expected 2028: Ph.D. in Planetary Sciences, Minor in Optical Sciences

Expected 2026: M.S. in Planetary Sciences

Thesis Advisor: Walt Harris; Minor Advisor: Daewook Kim

Princeton University, Princeton, NJ, 2020–2024

A.B., Honors in Astrophysical Sciences with Service Focus

Thesis Advisors: Chris Chyba, Katherine de Kleer, Maria Camarca

PEER-REVIEWED PUBLICATIONS (*indicates undergraduate advisee)

C. Meyer, J. Flores, J. Corliss, J. Alday, W. Harris. Experimental validation of a dual-bandpass spatial heterodyne spectrometer. *In prep.*

C. Meyer, J. Flores, H. Das*, H. Marchinek*, J. Corliss, E. Hamden, H. Chung, W. Harris. Spatial Heterodyne Interferometric Molecular Cloud Observer (SHIMCO) I. Target selection and modeling. *Submitted to AJ*.

C. Meyer, M. Camarca, K. de Kleer, A. Thelen, C. Chyba, B. Butler. Revealing Callisto's Near-subsurface Thermophysical Properties with ALMA Calibration Data. *Planetary Science Journal*, 7(1), 10. <https://doi.org/10.3847/PSJ/ade7ee>.

M. Camarca, K. de Kleer, B. Butler, A. Thelen, **C. Meyer**, A. Akins, I. de Pater, M. Gurwell (2025). A Multifrequency Global View of Callisto's Thermal Properties from ALMA. *Planetary Science Journal*, 6(8), 183. <https://doi.org/10.3847/PSJ/ade7ee>.

R. Bandyopadhyay, **C. Meyer**, W. Matthaeus, D. McComas, S. Cranmer, J. Halekas, J. Huang, D. Larson, R. Livi, A. Rahmati, P. Whittlesey, M. Stevens, J. Kasper, S. Bale (2023). Estimates of proton and electron heating rates extended to the near-Sun environment. *Astrophysical Journal Letters*, 955(2), L28. <https://doi.org/10.3847/2041-8213/acf85e>.

CONFERENCE PROCEEDINGS (*indicates undergraduate advisee)

C. Meyer, J. Flores, J. Corliss, W. Harris. Characterization of a symmetric-facet dual-ruled grating for spatial heterodyne spectroscopy. *Submitted to SPIE Proc.*

S. Agarwal, R. Maciel, J. Corliss, W. Harris, E. Hamden, **C. Meyer**, J. Alday, H. Marchinek*, J. Martinez Castillo, H. Das*, E. Garcia, N. Yescas, D. Truong (2025). Advancing remote sensing with SHEUVI: experimental validation of dual-ruled grating for extreme ultraviolet applications. *Proc. SPIE 13624, Astronomical Optics: Design, Manufacture, and Test of Space and Ground Systems V*, 136241S. <https://doi.org/10.1117/12.3063925>.

CONFERENCE ABSTRACTS/PRESENTATIONS

“Enabling technologies for the first dual-bandpass spatial heterodyne spectrometer”

Contributed talk: LPL Graduate Student Colloquium Day 2026, Tucson, AZ

“SHIMCO I. Target Selection and Modeling”

Invited talk: LPLC Department Conference 2025, Tucson AZ.

Contributed talk: 247th AAS Meeting 2026, Phoenix, AZ

Poster: Arizona Space Institute Symposium 2025, Tucson, AZ.

“Thermal Imagery of Callisto Using ALMA Calibrator Data”

Contributed talk: DPS Meeting 2024, Boise, ID.

Contributed talk: LPLC Department Conference 2024, Tucson, AZ.

3-minute video: Princeton Research Day 2024, Princeton, NJ.

Poster: 243rd AAS Meeting 2024, New Orleans, LA.

Contributed talk: Caltech SURF Symposium 2023. Pasadena, CA.

“OBserving Sodium Doublet in Io’s Aurorae (OBSiDIAn) Spectrograph”

Poster: AstroTech Spectrograph Symposium 2023. Berkeley, CA.

“Estimates of Heating Rates in the Near-Sun Environment”

3-minute video: Princeton Research Day 2023, Princeton, NJ.

“Automating Inspection of Attenuation Grids and Carbon Foils”

Contributed talk: Princeton USRP Symposium 2022. Princeton, NJ.

PUBLIC OUTREACH & SERVICE

Grants Reviewer, Graduate and Professional Student Council, 2025–present.

Organizer, Lunar and Planetary Laboratory Conference, 2025–present.

Chambliss Poster Competition Judge, 247th AAS Meeting, 2026.

Co-Founder & Co-Editor-in-Chief, Princeton Undergraduate Research Journal, 2023–2024.

Founder & President, Princeton Astronomy Club, 2021–2024.

OUTREACH TALKS

Space Drafts 116, Tucson, AZ. *“The Universe’s Barcode: What Color in Space Tells Us”*

CLASSROOM TEACHING EXPERIENCE

Co-Founder & Co-Director, Other Worlds Astrobiology Initiative, 2025–present.

Graduate Instructor, University of Arizona Sky School, 2025–present.

Observation Assistant, Planets in the Universe, Princeton University, Fall 2023.

PROFESSIONAL DEVELOPMENT

Participant, P.I. Launchpad, NASA Ames Research Center, 2025.

Participant, Code/Astro Summer School, Northwestern University, 2025.

Participant, AstroTech Summer School, UC Berkeley, 2023.

ACADEMIC AWARDS/FELLOWSHIPS

Shirley D. Curson Education Plus Travel Grant (\$1,000), University of Arizona, 2026.
College of Science Galileo Circle Scholarship, University of Arizona, 2026.
LPL Leif Erland Andersson Award for Service and Outreach, University of Arizona, 2026.
Arizona/NASA Space Grant Fellowship, University of Arizona, 2025.
Best Graduate Student Talk, Lunar and Planetary Laboratory Conference, 2024.
NSF Graduate Research Fellowship, 2024.
Outstanding Presentation Award, Princeton University, 2024.
OUR UG Fund for Academic Conferences (\$750), Princeton University, 2024.
Summer UG Research Fellowship, Caltech, 2023.
Outstanding Presentation Award, Princeton University, 2023.
UG Summer Research Program, Princeton University, 2022.
Service Focus Fellowship, Princeton University, 2021.
International Internship Program, Princeton University, 2021.
Service Leadership Fellowship, Princeton University, 2020.

RESEARCH/OUTREACH GRANTS

GPSC Research & Project (ReaP) Grant (\$1,350), University of Arizona, 2026.
Proposal Title: “*Seeing Both Ends of the Rainbow: A Novel Dual-Band Spectrometer for Line-Pair Diagnostics*”
Arizona Astrobiology Center Seed Grant (\$6,500), 2025.
Proposal Title: “*Other Worlds: A STEM Pathway for Institutionalized Youth*”

RESEARCH ADVISORSHIP

Elliot Emerson (undergraduate), 2025–present.
Heerok Das (Arizona/NASA Space Grant; undergraduate), 2024–present.
Emrick Little (STAR Lab; high school student), 2025–2026.
Hayden Marchinek (Arizona/NASA Space Grant; undergraduate), 2024–2025.
Jasmine Martinez Castillo (Arizona/NASA Space Grant; undergraduate), 2024–2025.